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Assessing Patient Exposure in PET/CT Scans: A Head and Neck Dose Parameter Study

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Introduction:

Positron Emission Tomography/Computed Tomography (PET/CT) plays a crucial role in diagnosing, staging, and monitoring head and neck malignancies. However, the combined radiation dose from both PET (e.g., 18F-FDG) and CT components raises significant concerns regarding patient exposure and long-term risks.

Primary Aim:

To evaluate patient radiation exposure during PET/CT head and neck procedures

Methodology:

Study Location:

The investigation was conducted at a PET/CT research center in Almana Hospital, Eastern Province, **Dammam, Saudi Arabia**. This facility serves as one of the country's largest referral hospitals and specialized oncology clinics.

Equipment Specifications:

Patient exposure was assessed using a combined **Siemens Biograph mCT Flow** system, renowned for its precision in nuclear medicine imaging.

Result:

A total of 58 Head and Neck PET-CT scans were analyzed. The mean age of the patients was 52.28 years (SD = 19.27). Patient BMI was distributed across categories, with no single category exceeding 50%. The mean DLP was 540.82 mGy.cm (SD: 192.34). The mean CTDI was 4.38 mGy (SD: 0.52). The mean radionuclide activity was 314.76 MBq (SD: 42.73).

Discussion:

Table 3 provides a comparative analysis of the mean injected radiotracer activity (MBq) used in our study against values reported in previous international studies. This comparison is crucial for benchmarking our local practices against global standards and identifying potential areas for optimization.

The mean activity of **314.4 MBq** administered in our current study aligns closely with the value of **323 MBq** reported by Martí-Climent et al. (2017). It falls within a reasonable range above the **280 MBq** reported by Chipiga et al. (2019). This suggests that the protocol used at our institution in Saudi Arabia is broadly consistent with current international practices for head and neck PET/CT imaging.

Conclusion:

These findings underscore the importance of optimizing dose parameters, such as radiotracer activity and CT protocols, to reduce patient exposure while maintaining diagnostic quality. Further research and protocol adjustments are recommended to minimize radiation risks without compromising clinical outcomes.

References:

Chipiga, L., Vodovatov, A., Zvonova, I., Poyda, M., & Bernhardsson, C. (2019). Assessment of patient doses and corresponding radiation risks from PET/CT examinations in the Russian Federation. AIP Conference Proceedings, 2089. https://doi.org/10.1063/1.5095737

Martí-Climent, J. M., Prieto, E., Morán, V., Sancho, L., Rodríguez-Fraile, M., Arbizu, J., García-Velloso, M. J., & Richter, J. A. (2017). Effective dose estimation for oncological and neurological PET/CT procedures. EJNMMI Research, 7(1). https://doi.org/10.1186/s13550-017-0272-5

Table 1: Demographic Data

Statistic	Age (Y)	Height (Cm)	Wight (Kg)	BMI
Mean	52.14285714	167.4285714	73.05	26.40392857
StDiv	19.7816119	9.829767984	13.8575424	5.868349089
Min	24	150	48	14.98
Max	92	180	105	34.17

Table 2: Dose Parameters

Statistic	Activity (MBq)	DLP (mGy.cm)	CTDI (mGy)	Scan Length (mm)	SSDE (mGy)	KVp
Mean	314.3678571	541.9360714	7.599285714	602.1428571	11.81142857	120
StDiv	43.85400065	197.5119697	0.386522277	373.5219478	3.494451384	0
Min	222	186.7	7.08	243	5.01	120
Max	395.9	761.99	8.57	1209	17.1	120